

I. Digital oscilloscopes

A digital oscilloscope is a device that converts an analog voltage input signal $V_{in}(t)$ into digital form, stores the waveform in memory, and displays the stored waveform. It can also make measurements and carry out other analysis functions on the waveform data.

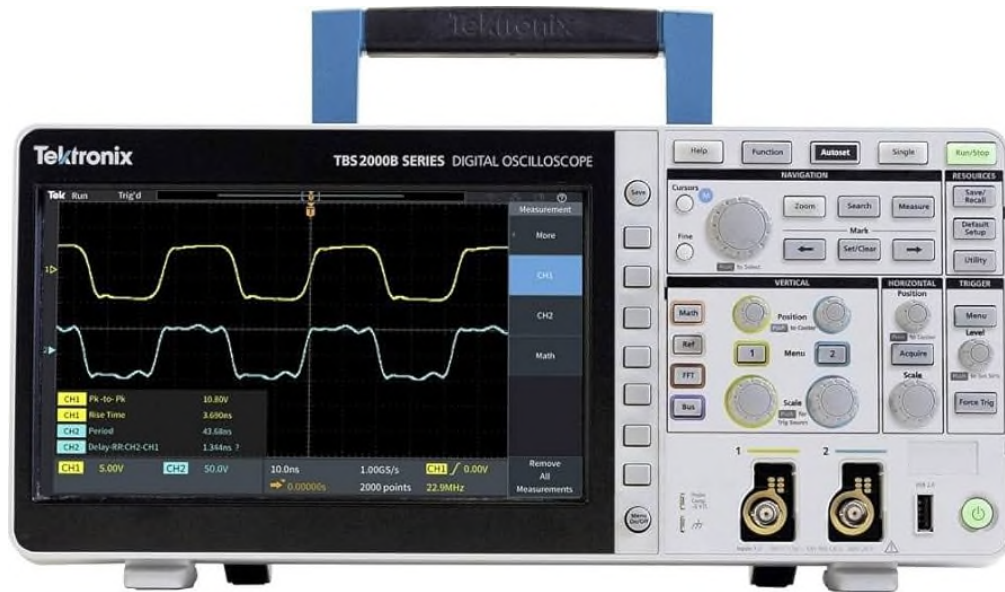


Figure 1. A Tektronix TBS2102B digital oscilloscope.

Figure 1 shows a typical digital oscilloscope, the Tektronix TBS2102B. It has two analog channels, which means that it can accept, process and display up to two analog signals at the same time. Two digitized waveforms are visible on the display in Figure 1.

A. Signal processing in a digital oscilloscope

Figure 2 shows a block diagram of a typical digital oscilloscope. The input signal first passes through an analog attenuator with adjustable attenuation. Following this, the signal passes through an analog offset circuit that can add or subtract a fixed voltage from the input signal. This allows the waveform to be moved up and down on the display. Next, the signal passes through an analog amplifier with adjustable gain. After this, the signal passes through a low-pass filter (typically with 20 MHz bandwidth) that can be switched in and out (not shown in Figure 2). This is useful when the signal has only lower frequency components, since switching in the filter can reduce the noise in the signal. Finally, the signal is passed to the heart of the oscilloscope, its analog-to-digital converter (ADC). The ADC converts a voltage in its input range $[V_{ADC,min},$

$V_{ADC,max}]$ to an integer expressed as a digital signal. Most often, this is an 8-bit number, so the oscilloscope distinguishes between $2^8 = 256$ different values of the voltage. This range of voltage values doesn't necessarily match the oscilloscope's input voltage range. For example, if the oscilloscope vertical sensitivity is set for 10 mV/div and has 10 vertical divisions, it is set up to display a voltage in a 100 mV range. The purpose of the input attenuator and input amplifier is to match this input voltage range to the voltage range of the ADC. This allows the oscilloscope to achieve its full vertical voltage resolution (typically 1 part in 256) for any setting of the input vertical sensitivity.

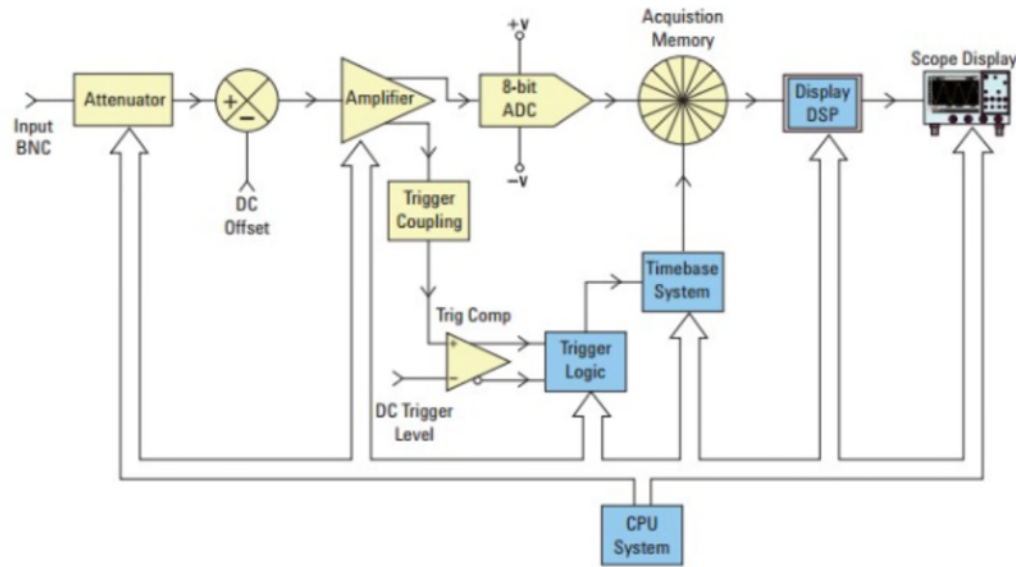


Figure 2. Block diagram of a digital oscilloscope. Figure is from <https://docs.keysight.com/kkbopen/how-an-oscilloscope-works-journey-to-the-center-of-the-scope-620139761.html>

The bandwidth B of the analog attenuator/amplifier is one of the most important specifications of the oscilloscope. For instance if you have a “100 MHz oscilloscope,” that means that the bandwidth of the input analog attenuator/amplifier is 100 MHz. This follows the usual definition of bandwidth as the -3dB point. That is, a sinusoidal signal with frequency B will be displayed with a voltage amplitude that is smaller than the input amplitude by a factor of $\sqrt{2}$ (i.e., factor of 2 lower in power, which is 3 dB lower in power.) Sine wave inputs of much lower frequency will be accurately displayed, while inputs of much higher frequency will be displayed with a highly reduced amplitude.

The ADC is set by the oscilloscope to record (*sample*) the input voltage $V_{in}(t)$ at a *sequence of times* $t_n = n\Delta t$, where $n = 0, 1, 2, \dots, N-1$ is an index that labels the samples, N is the total number of samples, and Δt is the time interval between adjacent samples, as illustrated in Figure 3.

$f_s = \frac{1}{\Delta t}$ is the *sampling rate*. The total length of the recorded signal is $T = (N - 1)\Delta t$. The values of Δt , f_s , N , and T depend on the oscilloscope settings.

As the input signal is sampled, it is transferred to the *acquisition memory*. The start and stop time of the sampling is controlled by the *timebase system*. This is normally controlled by triggering of the oscilloscope. Triggering allows successive waveforms to be synchronized in time on the oscilloscope display. Once a recording has stopped, the waveform is transferred to a digital signal processor for further processing and output out the oscilloscope display.

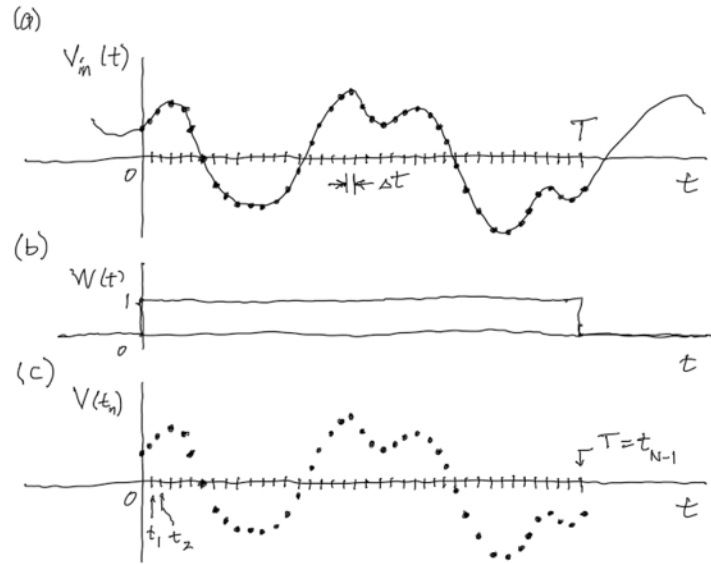


Figure 3. (a) Input signal $V_{in}(t)$ to a digital oscilloscope, showing also the recorded samples. (b) Window function $W(t)$ for the recording. (c) Sampled signal $V(t_n)$ recorded by the oscilloscope.

Another important oscilloscope specification is its *maximum sampling rate* $f_{s,max}$. This should be at least 2.5 times the analog bandwidth for reasons we'll discuss below. One more important specification has to do with digital signal processing times. Once a waveform recording is completed, there is a time delay before the start of the next waveform recording while the scope completes the digital processing on the previous waveform. This delay can be much longer than T . The *maximum waveform capture rate* is the number of waveforms per second that can be captured, as limited by signal processing times. In a low-end oscilloscope this could be just a few thousand waveforms per second. In a higher-end scope it could be 500,000 waveforms per second or more.

Waveform capture rate is not very important if the signal repeats exactly. However if the signal fluctuates and has some occasional rare glitches, waveform capture rate is very important. A scope with a low capture rate can completely miss the rare events, while one with a high rate will see them.

The display update rate is typically much slower than the waveform capture rate. In this case many waveforms are captured for each update of the display. In most oscilloscopes you can choose what you want to display. For instance you can display just the first captured waveform, the average of the captured waveforms, or all of the waveforms in an overlapping display. The last display option is useful if your input signal has fluctuations.

The signal $V(t)$ recorded by an oscilloscope differs from the true input signal $V_{in}(t)$ in several important ways. First, only a finite length sample of duration T of the signal is recorded. You can think of this as the result of the multiplication of the true input signal $V_{in}(t)$ by a *window function* $W(t)$ as shown in Figure 3, *i.e.* $V(t) = W(t)V_{in}(t)$. Also, the signal $V(t_n)$ is recorded only over a finite set of times $\{t_n\}$ rather than continuously. As we'll see later, in some cases this is a significant disadvantage while in other cases it is not a disadvantage at all. Finally, in a digital oscilloscope the signal is recorded as integers with accuracy limited by the number of bits. Lower end scopes, including the Tektronix TBS1052C oscilloscopes used in this class, record voltage samples as 8-bit numbers with rather poor vertical resolution of 1 part in 256.

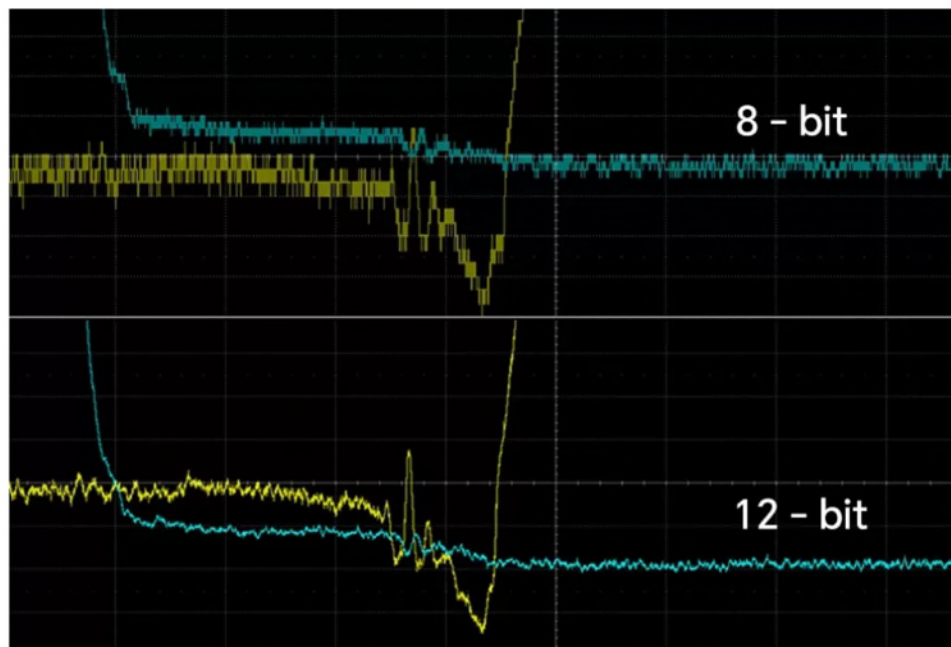


Figure 4. Comparison of display of waveforms on an 8-bit and a 12-bit oscilloscope.

Mid-range and higher-end oscilloscopes have 12 bit (1 part in 4096) up to 16 bit (1 part in 65,536) vertical resolution. The difference is especially noticeable with some vertical magnification of the signal as shown in Figure 4. For example in the upper yellow trace for an 8-bit oscilloscope, you can clearly see quantization of the recorded input voltage in units of the least significant bit. Each unit is $1/256 = 0.39\%$ of the full vertical voltage range. The steps appear larger than that in Figure 4 because the vertical scale has been magnified. In contrast, the

recorded signal in the 12-bit oscilloscope shows no obvious quantization of the same signal viewed at the same scale.

B. Summary of key digital oscilloscope specifications

- Analog bandwidth B .
- Maximum sampling rate $f_{s,MAX}$.
- Number of analog channels. This is the number of analog inputs that can be recorded simultaneously.
- Vertical resolution. This is the number of bits that are used to record the voltage signals.
- Waveform capture rate.
- Memory depth. This is the number of data points that can be stored in the oscilloscope memory. A greater memory depth means that the oscilloscope can record data at any given sampling rate for a longer time. However a larger memory depth can also slow down the waveform capture rate since it increases the numerical processing required for a capture.

Below I've given a comparison of the specifications of two oscilloscopes: the Tektronix TBS1052C oscilloscope used in our lab, and the higher-end Tektronix MSO-44B-BW-1000. As you can see, prices rise dramatically for scopes with very high bandwidth, sampling rate, and waveform capture rate. If you want see a really expensive oscilloscope, [look at the the Keysight UXR1104B Infiniium 110 GHz 4 channel oscilloscope](#). It costs over \$2M.

Oscilloscope	Price	Number of analog channels	Analog bandwidth B	Maximum sampling rate $f_{s,MAX}$	Vertical resolution	Waveform capture rate	Memory depth
Tektronix TBS-1052C	\$622	2	50 MHz	1 Gs/s	8 bits	up to 5,000 waveforms/s	20 kpts
Tektronix MSO-44B-BW-1000.	\$25,800	4	1 GHz	6.25 Gs/s	12 bits	>500,000 waveforms/s	32.5 Mpts

II. Spectral Analysis

A. Discrete Fourier Transform (DFT)

One of the more useful features of digital oscilloscopes is their capability to calculate and plot Fourier transforms of a captured waveform. Since the captured waveform is discrete rather than continuous, they apply a Discrete Fourier Transform (DFT) to the data. Here, we'll lay out the basics of the DFT.

We start with a recorded waveform $V(t_n)$, with $t_n = n\Delta t$ and $n = 0, 1, \dots, N-1$. We'll assume here that N is an even number, since that is almost always the case in practice.

The *Discrete Fourier Transform* of V is

$$\tilde{V}(f_k) = \frac{1}{N} \sum_{n=0}^{N-1} V(t_n) e^{-\frac{j2\pi kn}{N}} \quad (1)$$

The *Inverse Discrete Fourier Transform* (IDFT) is

$$V(t_n) = \sum_{k=0}^{N-1} \tilde{V}(f_k) e^{+\frac{j2\pi kn}{N}} \quad (2)$$

The DFT $\tilde{V}(f_k)$ is also discrete. It is a function of frequencies $f_k = k\Delta f$, with $k = 0, 1, 2, \dots, N-1$, and $\Delta f = \frac{1}{N\Delta t}$ the spacing between adjacent frequency points. Note that with sampling time interval Δt , the highest frequency component the sampled signal can contain is $\sim 1/\Delta t$.

$\Delta f = \frac{1}{N\Delta t}$ divides this maximum frequency into N equal intervals.

You will often see these equations written with the factor of $1/N$ in Equation (2) rather than Equation (1). Both versions are mathematically correct, but I prefer the version above because in that version the variables $\tilde{V}(f_k)$ are more directly related to the amplitudes of the component sine waves.

Mathematically, both $V(t_n)$ and $\tilde{V}(f_k)$ in Equations (1) and (2) are complex functions. So, both are determined by $2N$ real numbers: N for each real part, and N for each complex part. However since $V(t_n)$ is real in practice, all of the numbers specifying its imaginary part are zero. So we are left with a seeming puzzle: how can the N nonzero real numbers specifying $V(t_n)$ be mathematically sufficient to determine the $2N$ real numbers specifying $\tilde{V}(f_k)$? Part of the answer is that the condition that $V(t_n)$ is real places restrictions on $\tilde{V}(f_k)$. Consider $k = N - i$. We then have

$$\tilde{V}(f_{N-i}) = \frac{1}{N} \sum_{n=0}^{N-1} V(t_n) e^{-\frac{j2\pi(N-i)n}{N}} = \frac{1}{N} \sum_{n=0}^{N-1} V(t_n) e^{-j2\pi n} e^{+\frac{j2\pi in}{N}} = \frac{1}{N} \sum_{n=0}^{N-1} V^*(t_n) e^{+\frac{j2\pi in}{N}} = \tilde{V}^*(f_i) \quad (3)$$

To simplify

$$\tilde{V}(f_{N-i}) = \tilde{V}^*(f_i) \quad (4)$$

That is, the transform amplitude $\tilde{V}(f_{N-1})$ contains the same information as $\tilde{V}(f_1)$, the transform amplitude $\tilde{V}(f_{N-2})$ contains the same information as $\tilde{V}(f_2)$, and so on. It follows

that the complete waveform information is specified by the set

$\left\{ \tilde{V}(f_0), \tilde{V}(f_1), \tilde{V}(f_2), \dots, \tilde{V}\left(f_{\frac{N}{2}}\right) \right\}$, which contains $\frac{N}{2} + 1$ complex variables, or equivalently $2\left(\frac{N}{2} + 1\right) = N + 2$ real quantities. That's better, but it is still 2 more than the number of real variables describing $V(t_n)$. However, note that

$$\text{Im}(\tilde{V}(f_0)) = \text{Im}\left(\frac{1}{N} \sum_{n=0}^{N-1} V(t_n) e^{-j0}\right) = 0 \quad (5)$$

and

$$\text{Im}(\tilde{V}(f_{N/2})) = \text{Im}\left(\frac{1}{N} \sum_{n=0}^{N-1} V(t_n) e^{-j\pi kn}\right) = 0 \quad (6)$$

since every term in the sums is real. Thus out of our $N + 2$ variables specifying $\tilde{V}(f_N)$, two of them are zero, leaving a total of exactly N non-zero real variables for the full specification of $\tilde{V}(f_N)$. This resolves the puzzle, as summarized in Figure 5 for the case $N = 8$.

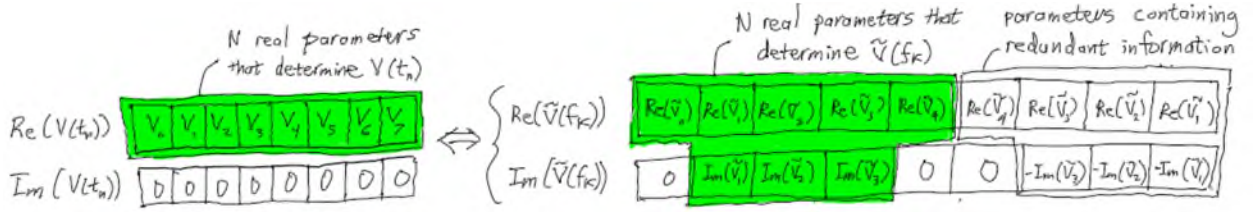


Figure 5. Illustration of zero, non-zero, and redundant variables for a DFT with $N = 8$. The non-zero, nonredundant quantities are highlighted in green.

Although the redundant parameters do not carry any new information, it is still important to include them in any calculation with Equations (1) or (2), since that is required to obtain quantitatively correct results.

The basic idea of Fourier analysis is that a signal $V(t)$ can be expressed as a summation of sine and cosine waves of different frequencies. We can see this explicitly by writing out Equation (2) in terms of its real and imaginary parts:

$$\begin{aligned} V(t_n) &= \sum_{k=0}^{N-1} \tilde{V}(f_k) e^{+j\frac{2\pi kn}{N}} = \sum_{k=0}^{N-1} \tilde{V}(f_k) \cos\left(\frac{2\pi kn}{N}\right) + j \sum_{k=0}^{N-1} \tilde{V}(f_k) \sin\left(\frac{2\pi kn}{N}\right) \\ &= \sum_{k=0}^{N-1} \tilde{V}(f_k) \cos(2\pi f_k t_n) + j \sum_{k=0}^{N-1} \tilde{V}(f_k) \sin(2\pi f_k t_n) \end{aligned} \quad (7)$$

Equation (7), the IDFT, expresses the sampled signal $V(t_n)$ as a linear combination of cosine waves $\cos(2\pi f_k t_n)$ and sine waves $\sin(2\pi f_k t_n)$ of different frequencies f_k . The quantity $\tilde{V}(f_k)$ gives the amplitude of the f_k “frequency component.” The condition $\tilde{V}(f_{N-i}) = \tilde{V}^*(f_i)$ ensures that $V(t_n)$ is a real quantity.

Since $\tilde{V}(f_k)$ is complex, it takes two real functions to specify it. One way is to specify its real and imaginary parts, from which we can construct $\tilde{V}(f_k) = \text{Re}(\tilde{V}(f_k)) + j \text{Im}(\tilde{V}(f_k))$. The other way is to specify its magnitude and phase, from which we can construct $\tilde{V}(f_k) = |\tilde{V}(f_k)| e^{j\phi(\tilde{V}(f_k))}$.

$|\tilde{V}(f_k)|$ is the *amplitude spectrum* of $V(t_n)$ and $\phi(\tilde{V}(f_k))$ is the *phase spectrum* of $V(t_n)$.

Some oscilloscopes can display these quantities, while others display only the Fourier power spectrum discussed below.

B. Fast Fourier transform (FFT)

Although Equations (1) and (2) are relatively straightforward linear equations, their solution with standard methods can become very computationally intensive if the number of data points becomes very large. Fortunately, there exists a fast algorithm for solving these equations called the *Fast Fourier Transform* (FFT). I won’t cover how it works here, but it is so fast that modern oscilloscopes can compute the DFT and display it in real time as data is coming in.

C. Sample FFT calculation with Python

The FFT function is available in data analysis packages as well. Below is a program that shows an example of how to calculate the FFT with Python. To keep things simple, we’ll start with the input signal $V_{in}(t) = A \sin(2\pi f_0 t + \phi)$.

```
import matplotlib.pyplot as plt
import numpy as np
from scipy.fftpack import fft, ifft

# total measurement time
T = 0.001
# number of samples
N = 200
# sampling interval
dt = T/N
# array of times tn
tn = np.arange(0,T,dt)

# sampled function V(t)
```

```

f0 = 20315
A = 1.
phi = 0.22
def V(t):
    return A*np.sin(2*np.pi*f0*t+phi)

# print out values of parameters for convenient reference
print('Measurement time T = ' + str(T) + ' s')
print('Number of samples N = ' + str(N))
print('Sample interval \u0394t = ' + str(dt) + ' s')
print('Sampling rate fs = ' + str(f"{1/dt:,.1f}") + ' samples/s')
print('Input sine wave frequency f0 = ' + str(f0) + ' Hz')
print('Input sine wave amplitude A = ' + str(A) + ' V')
print('Input sine wave phase \u03c6 = ' + str(phi) + ' rad')

# quasi-continuous array of times for plotting V(t)
tqc = np.arange(0,T,dt/20)

# plot out the input and sampled signal
plt.figure()
# plot out quasicontinuous V showing input signal
plt.plot(tqc,V(tqc))
# plot out sampled signal V(tn)
plt.scatter(tn, V(tn), s = 10)
plt.xlabel('time (s)')
plt.ylabel('Signal V(t)')
plt.show()

# Calculate the FFT of the sampled signal
# Use normalization to match class notes
Vk = fft(V(tn))/N
# define array of frequencies fk to use for plotting
k = np.arange(N)
fk = k/(N*dt)

# Plot out real and imaginary part of the FFT
plt.scatter(fk, np.real(Vk), s = 10, c = 'b', label = 'Re(Vk)')
plt.scatter(fk, np.imag(Vk), s = 10, c = 'r', label = 'Im(Vk)')
plt.xlabel('Frequency (Hz)')
plt.ylabel('Fourier amplitude')
plt.legend()
plt.show()

# Plot out the magnitude of the FFT
plt.scatter(fk, np.abs(Vk), s = 10)

```

```

plt.xlabel('Frequency (Hz)')
plt.ylabel('Fourier amplitude magnitude |Vk|')
plt.show()

# Calculate inverse FFT
# use normalization to match class notes
IVn = N*ifft(Vk)

# plot out inverse FFT on top of input V(t)
# Result should match Vn plotted out above
plt.scatter(tn, IVn, s = 10, label = 'inverse Fourier transform')
plt.plot(tqc, V(tqc), label = 'input V(t)')
plt.xlabel('time (s)')
plt.ylabel('input V(t), IFFT')
plt.legend()
plt.show()

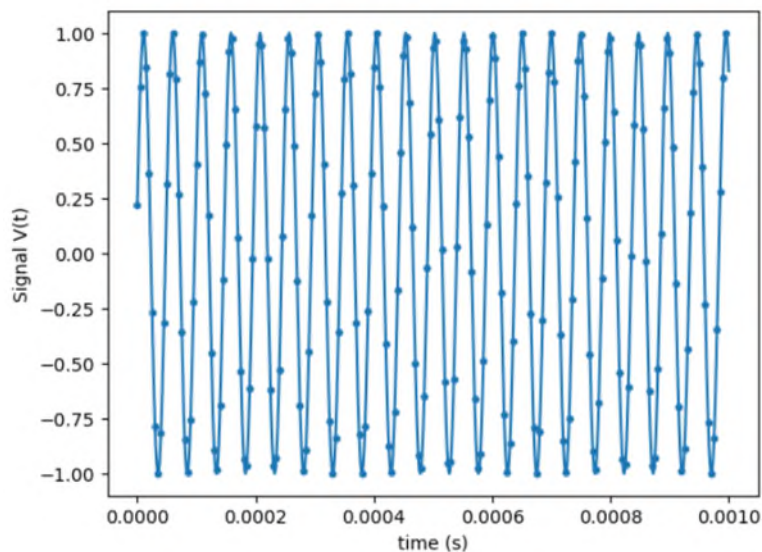
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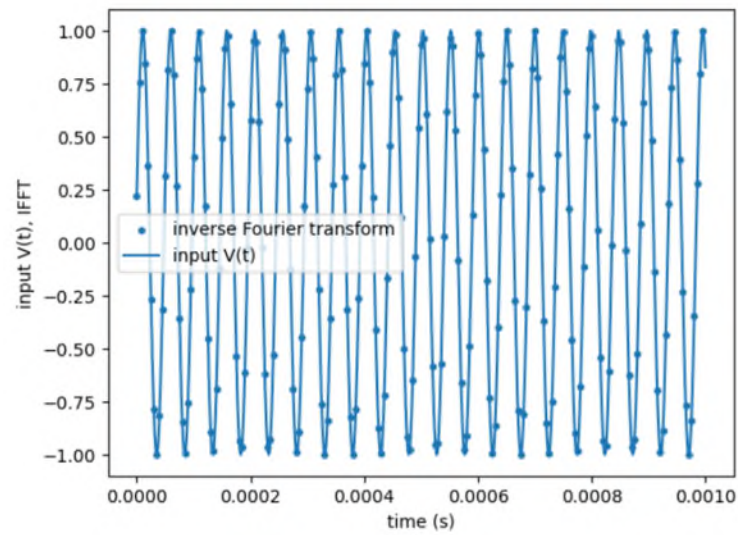
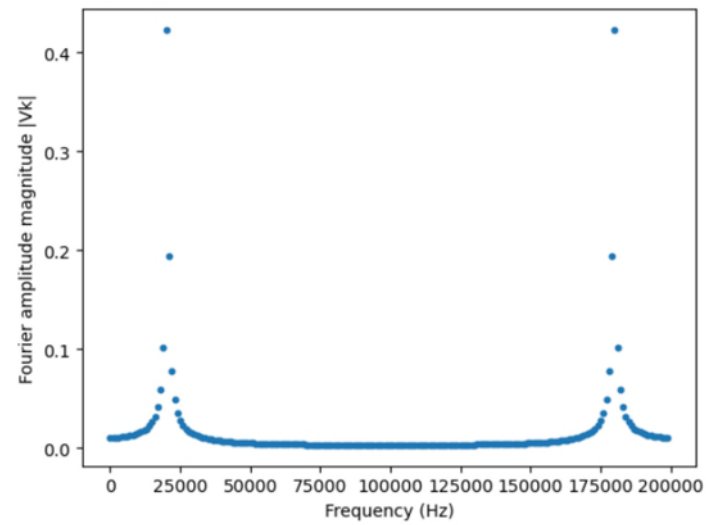
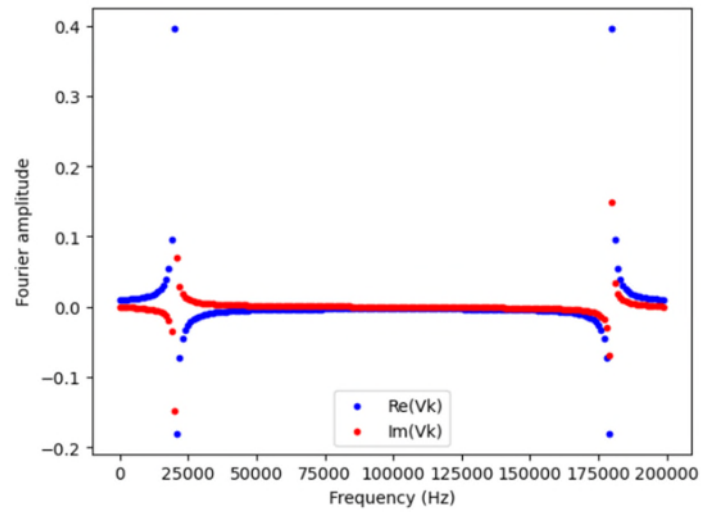
Here is the output of the program:

```

Measurement time T = 0.001 s
Number of samples N = 200
Sample interval Δt = 5e-06 s
Sampling rate fs = 200,000.0 samples/s
Input sine wave frequency f0 = 20315 Hz
Input sine wave amplitude A = 1.0 V
Input sine wave phase φ = 0.22 rad

```





Some things to notice about the result:

- In the first plot we show the input sine wave $V_{in}(t)$ as a continuous curve, and we show the sampled data as discrete points.
- We have chosen a sine wave frequency f_0 that is not an integer multiple of $1/T$. This is to avoid a special case which we'll discuss below.
- The DFT of the sampled data is shown in the second plot. Since it is a complex quantity, to fully specify the DFT we have to plot out both its real and imaginary part.
- The frequency span of the DFT is equal to the sampling rate.
- $\tilde{V}(f_{N-i}) = \tilde{V}^*(f_i)$. The DFT in the second half of the frequency span contains the same information as the first half.
- Often we only care about the magnitude of the DFT $|\tilde{V}(f_k)|$. That is shown in the third plot. As might be expected it shows a peak at $f_k = f_0$. A second peak at $f_k = f_N - f_0$ appears because of the condition $\tilde{V}(f_{N-i}) = \tilde{V}^*(f_i)$.
- You might naively have expected the DFT to just have a sharp spike at frequency f_0 . It does show a peak at frequency f_0 , but the peak has smooth tails on both sides. This is due to the fact that the DFT is the Fourier transform of a sine wave *modulated by the rectangular window function* $W(t)$, not the transform of an infinitely long sine wave.
- The last plot shows the Inverse DFT (IDFT) of the DFT on top of the input signal $V(t)$. The IDFT is identical to the original sampled data. This is a good check that the calculations are correct.

D. Single vs. double-sided Fourier spectrum

If we wish, we can define $\tilde{V}(f_k)$ on the frequency domain $\left[-\left(\frac{N}{2}-1\right)\Delta f, +\frac{N}{2}\Delta f\right]$ rather than the domain $[0, (N-1)\Delta f]$. The Fourier amplitudes at negative frequency are given by

$$\tilde{V}(f_{-k}) = \frac{1}{N} \sum_{n=0}^{N-1} V(t_n) e^{+j\frac{2\pi kn}{N}} = \tilde{V}(f_k)^* = \tilde{V}(f_{N-k}) \quad (8)$$

where the second equality follows from Equation (4). From this, we see that *the Fourier amplitudes for negative frequencies can be obtained by shifting the frequency components*

$\left[f_{\frac{N}{2}+1}, f_{N-1}\right]$ *in the original spectrum N places to the left.* This is illustrated in Figure 6 for the above Python FFT example. It can be done simply in Python with the Scipy instruction `fftshift()`. The resulting spectrum is referred to as a *double-sided Fourier spectrum*. The original spectrum

$\tilde{V}(f_k)$ is also considered a double-sided spectrum since the negative frequency components are specified in the second half of the spectrum.

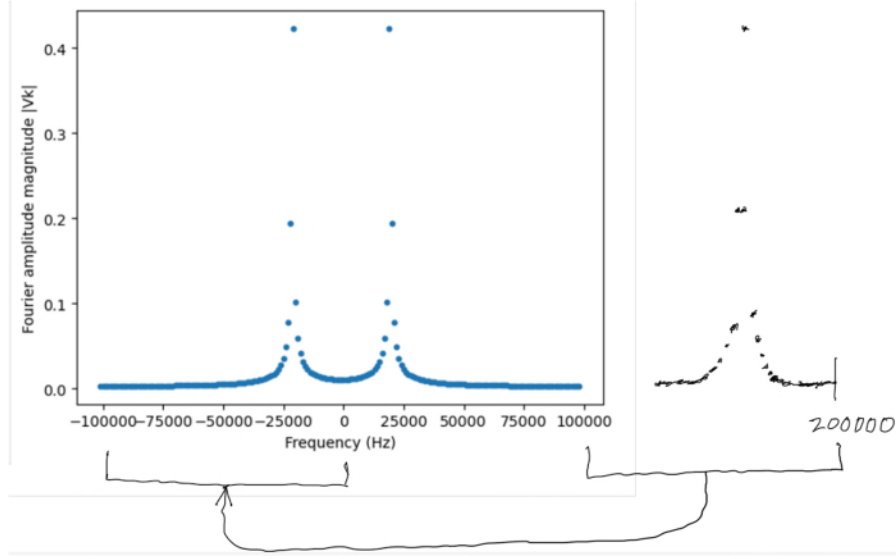


Figure 6. Magnitude of the double-sided FFT. It is constructed from the spectrum defined by Equation (1) by shifting the right half of the spectrum to negative frequencies.

The double-sided spectrum has the property $\tilde{V}_k^* = \tilde{V}_{-k}$. A spectrum with this property is said to be *conjugate symmetric*.

Since the negative frequency components carry redundant information to the positive frequency components, we often work with the positive frequency components only. This is referred to as a *single-sided Fourier spectrum*. It is defined by

$$\tilde{V}_{ss}(f_k) = \begin{cases} \tilde{V}(f_k), & \text{if } k = 0 \\ 2\tilde{V}(f_k), & \text{if } k = 1, 2, \dots, \text{ or } N/2 \end{cases} \quad (9)$$

The frequency domain of $\tilde{V}_{ss}(f_k)$ is $\left[0, \frac{N}{2} \Delta f_k\right]$.

Figure 7 shows the magnitude $|\tilde{V}_{ss}(f_k)|$ of the single sided FFT for the example of Section II.C.

This spectrum shows only the single peak at frequency f_0 . Most oscilloscopes only display single-sided spectra similar to the one in Figure 7.

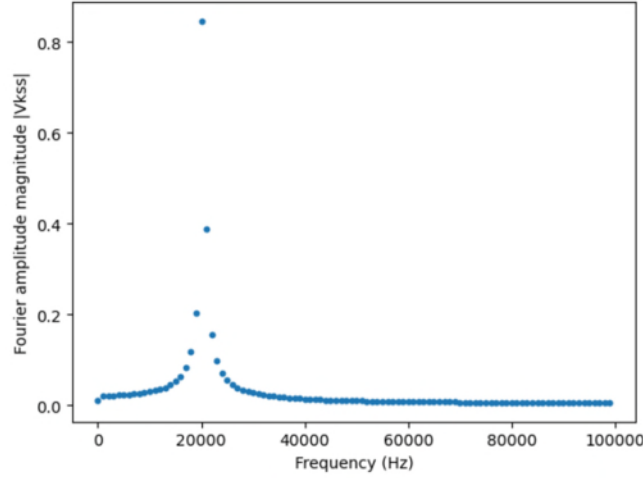


Figure 7. Magnitude of the single-sided DFT of the example function of Section II.C.

The reason for the factor of 2 in Equation (9) is that it makes the peak height in the Fourier spectrum identical to the amplitude of the input (or nearly so) for a case of a single frequency sine wave input. You can see this in Figure 7: the amplitude of the input sine wave was 1 Volt, and the magnitude of the peak at frequency f_0 is also very close to 1 Volt. (The small difference from 1 Volt will be discussed below.) This happens because each of the terms $\tilde{V}(f_k)$ and $\tilde{V}(f_{N-k})$ (for $k \neq 0$) contributes equally to the amplitude of the f_k frequency component in the sum in Equation (1). By contrast, the frequency component $k_0 = 0$ is included only once in the sum so we don't need the factor of 2 for that. The signal from the $k = 0$ component is

$$V(t_n) = \tilde{V}(f_0) \cos(0) + j\tilde{V}(f_0) \sin(0) = \tilde{V}(f_0) \quad (10)$$

That is, it is the DC component of the signal.

E. Nyquist-Shannon sampling theorem and reconstruction of continuous waveforms by interpolation

The Nyquist-Shannon sampling theorem states that:

If the Fourier transform of a signal $V(t)$ has zero amplitude for all frequencies above B Hertz, then $V(t)$ can be exactly determined from a stream of samples at sampling rate $2B$.

In other words, no information is lost by sampling as long as the sampling rate is greater than twice the largest frequency non-zero Fourier component of the signal. The required sampling rate $2B$ for this condition is called the *Nyquist rate*. The *Nyquist frequency* $f_s/2$ is the highest frequency Fourier component that can be reliably reconstructed from data sampled at rate f_s .

This principle is important for digital audio. Since the human ear cannot sense frequencies much above 20 kHz, this means that no audible information is lost with a digital audio recording as long as the sampling rate is larger than about 40 kHz.

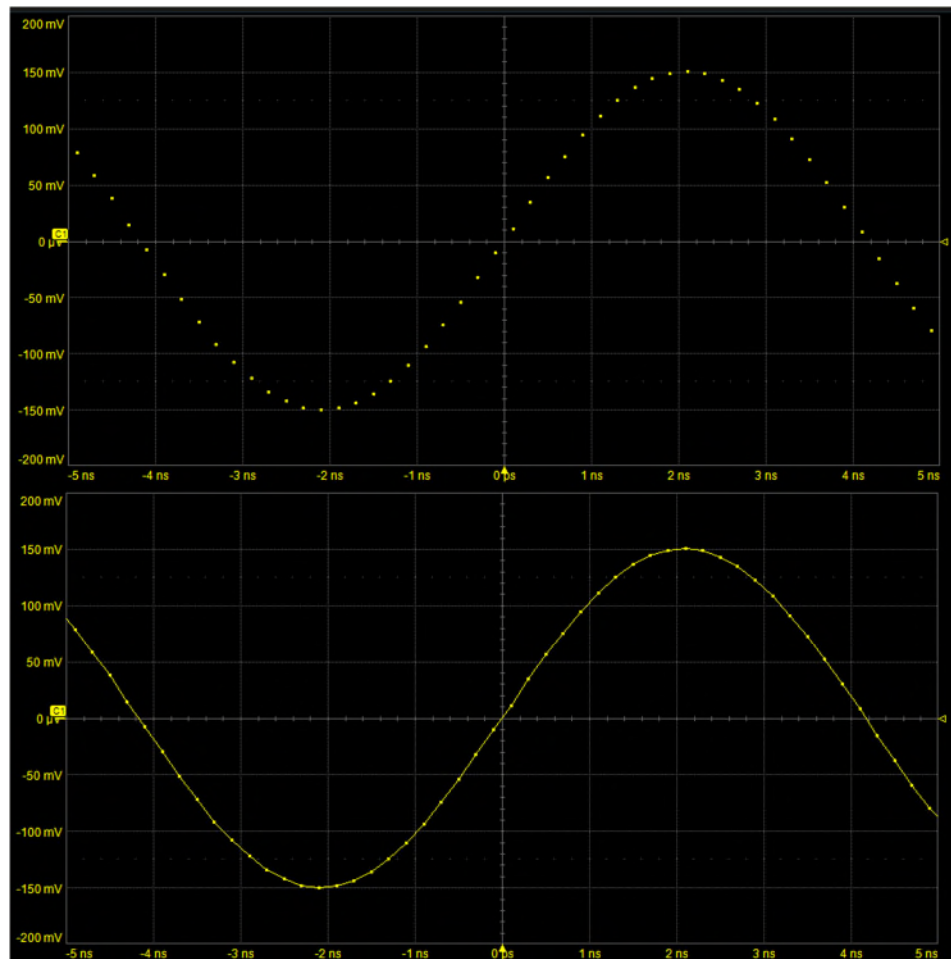


Figure 8. Oscilloscope interpolation. Upper trace: dots display of data recorded by oscilloscope. Lower trace: display showing interpolated data. The interpolated points fill in the gaps between the real sample points which are highlighted as intensified dots. Figure is from <https://www.embedded.com/filling-in-the-blanks-in-digital-oscilloscope-waveforms/>

This principle is also used in digital oscilloscopes to reconstruct a continuous waveform from the sampled waveform. To do this, the oscilloscope interpolates between the sampled data points, as shown in Figure 19. You can usually select between a linear interpolation and a $\sin(x)/x$ interpolation. In principle, the $\sin(x)/x$ interpolation reconstructs the continuous waveform exactly provided that the sampling rate is larger than the Nyquist rate. By default, the oscilloscope displays the interpolated waveform, as shown in the lower trace of Figure 19. The interpolated waveform is also used for automatic measurements, which improves the measurement accuracy. You can also opt to see the non-interpolated data points by selecting a “dots” display, as shown in the upper trace in Figure 19.

F. Aliasing

Aliasing is a flaw in the display of sampled data that can occur when the sampling rate is lower than the Nyquist rate. In this case, substantial information on the true shape of the waveform is lost, and the displayed waveform may be grossly inaccurate. A classic example is that of a sine wave that is sampled at a rate below the Nyquist rate. This can result in the display of a “sine wave” that has a much lower frequency than the true signal, as shown in Figure 20.

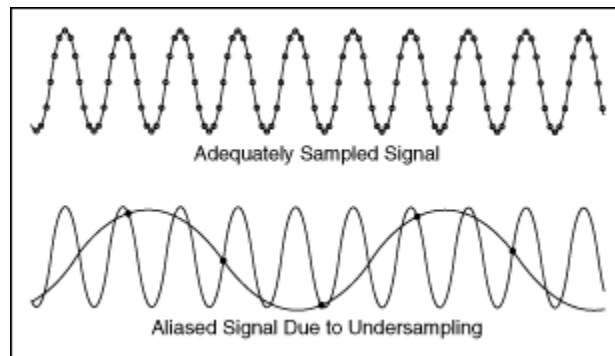


Figure 9. Aliasing of an under-sampled sine wave. Figure is from a [National Instruments article about aliasing](#).

Aliasing can also give rise to incorrect features in a Fourier spectrum, as illustrated in Figure 21. What happens is that Fourier components above the Nyquist frequency are “reflected” past the right edge of the spectrum so that they appear as false frequency components in the frequency range $[0, f_s / 2]$.

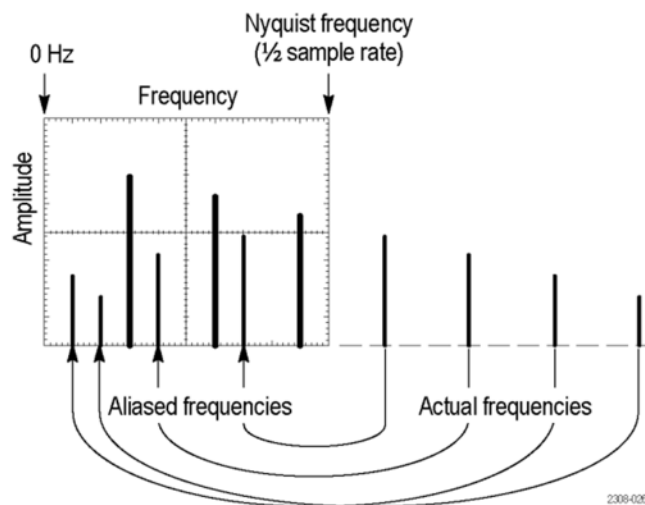


Figure 10. Effect of aliasing on Fourier spectra. Figure is from the [TBS2000B series oscilloscope manual](#).

To prevent this effect from occurring, it is important that the signal not contain any significant Fourier components above the Nyquist frequency. In the event that aliasing is observed in a Fourier spectrum, it can be eliminated by either increasing the sampling rate so that the Nyquist

frequency is above any signal Fourier components. Alternatively the signal can be filtered so as to eliminate the Fourier components above the Nyquist frequency.

G. Fourier Power spectrum

The *Plancherel theorem*, a special case of *Parserval's theorem*, states that

$$\frac{1}{N} \sum_{n=0}^{N-1} |V(t_n)|^2 = \sum_{k=0}^{N-1} |\tilde{V}(f_k)|^2 \quad (11)$$

Now, suppose that the voltage $V(t)$ is driving a load with resistance R_L . At any given time, the

power delivered to the load is $P(t_n) = \frac{|V(t_n)|^2}{R_L}$. Thus $\frac{1}{R_L} \frac{1}{N} \sum_{n=0}^{N-1} |V(t_n)|^2$ is the *average power* $\langle P \rangle$ delivered to the load during the time interval T .

It is conventional in electronics work to assume that $R_L = 1 \Omega$ in FFT formulas for power. With this convention the average power is

$$\langle P \rangle = \sum_{k=0}^{N-1} |\tilde{V}(f_k)|^2 = |\tilde{V}(f_0)|^2 + 2 \sum_{k=1}^{N/2} |\tilde{V}(f_k)|^2 = \sum_{k=0}^{N/2} S(f_k) \quad (12)$$

where

$$S(f_k) = \begin{cases} |\tilde{V}(f_k)|^2, & \text{if } k = 0 \\ 2|\tilde{V}(f_k)|^2, & \text{if } k \neq 0 \end{cases} \quad (13)$$

Note that $S(f_k)$ is defined only over the frequency range $\left[0, \frac{N}{2} \Delta f\right]$ rather than the full range $[0, (N-1)\Delta f]$, i.e. $S(f_k)$ is single-sided. The second equality in Equation (12) follows from Equation (4).

Each term in the sum in Equation (12) can be viewed as a contribution $S(f_k)$ to the *average power* $\langle P \rangle$ which comes from the frequency component f_k . We refer to $S(f_k)$ as the *power spectrum*. Its units are V^2 , with the tacit understanding that this must be divided by the load resistance to convert the units to Watts. Figure 11 shows the calculated power spectrum for the example of section II.C. Note that the peak power, at frequency f_0 is 0.5 W (or nearly so). That matches the time-averaged power $\langle P \rangle = (1 \text{ Volt})^2 \langle \sin^2(2\pi f_0 t + \phi) \rangle = 0.5(1 \text{ Volt})^2$.

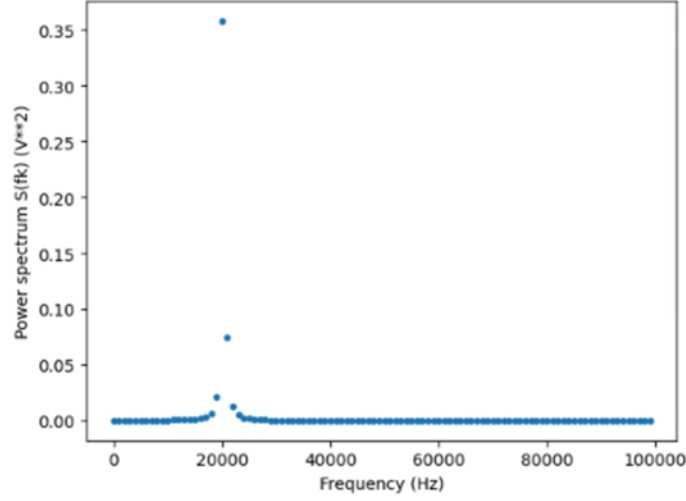


Figure 11. Power spectrum $S(f_k)$ for the example signal of section II.C.

The power spectrum is usually displayed on a *decibel (dB) scale*. To do this, we have to define a reference power. A common choice is the *decibel-Volt (dBV)* (sometimes dBV-rms for clarity). This is defined as

$$\# \text{dBV} = 10 \log_{10} \left(\frac{V_{rms}^2}{(1 \text{ V rms})^2} \right) = 20 \log_{10} \left(\frac{V_{rms}}{1 \text{ V rms}} \right) \quad (14)$$

Another common choice is *decibel-milliwatts (dBm) into 50 Ohm load*. This corresponds to a reference voltage of 0.22 Volts rms. Also note that for a single Fourier component of the power spectrum $V_{k,rms}^2 = S_k$, since S_k is already in units rms-voltage-squared. That is

$$\# \text{dBV} = 10 \log_{10} (S_k) \quad (15)$$

This is shown in Figure 12 for the example of section II.C.

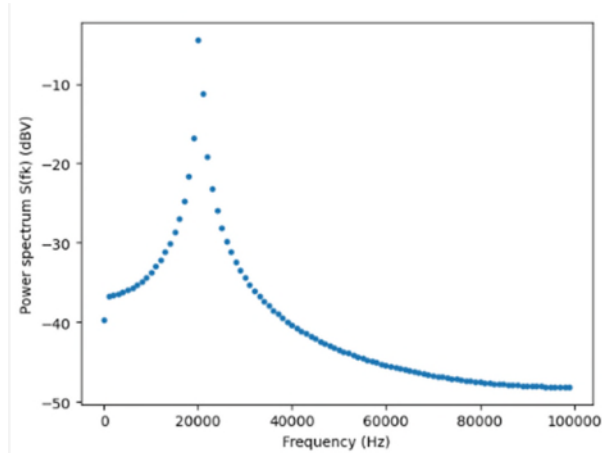


Figure 12. Plot of the power spectrum of the example of section II.C on a dBV scale.

H. Windowing effects

If you're familiar with Fourier transforms, you might remember that the Fourier transform of a sine wave modulated by a rectangular window function is a sinc function that has oscillations in it. If that is the case, then why aren't such oscillations visible in the above examples of Fourier spectra? The answer is that this happens because the spacing between the discrete frequency points is too large. We can see that this is the case by *padding* our input data with following zeros. That is, we take

$$V_{pad}(t_n) = \begin{cases} V(t), & \text{if } 0 < n < N-1 \\ 0, & \text{if } N-1 < n < N_{pad}-1 \end{cases} \quad (16)$$

where $N_{pad} < N$. The non-zero signal in V_{pad} is exactly the same as that in V . But, because the time range of the signal $T_{pad} = (N_{pad} - 1)\Delta t$ is greater than T , the frequency interval Δf_{pad} is smaller than Δf .

Figure 13 shows a comparison of the DFT $V(f_k)$ of the example of section II.C with the DFT of the same function $V(t)$ except that it has been padded with zeros up to a total time of $10T$. The DFT of the padded signal is the same except that it has higher frequency resolution. We see that the DFT of the padded signal does show a sinc function as expected. Also we can see from the figure the explanation of the lack of oscillation in the above examples of a DFT: *the spacing of the frequency points in those DFTs is the same as the spacing of the oscillations in the sinc function*. We also see the explanation for the imperfect agreement between the amplitude of the DFT and the expected amplitude of 0.5: *the highest point in the DFT may be a little ways down the side of the peak in the sinc function, which causes it to have an amplitude smaller than the true peak amplitude*.

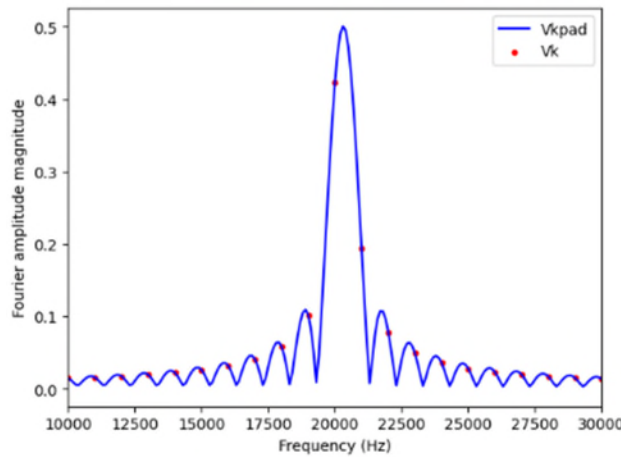


Figure 13. Comparison of DFT V_k of the example of section II.C with the DFT of the same signal padded by zeros up to a total time interval of $10T$.

The rectangular window function has a number of undesirable effects on the displayed DFT, as follows:

- The attenuation of the highest Fourier amplitude relative to the true maximum amplitude is one. Also,
- The shape of the curve can be strongly affected by the exact way that the input function is cut-off by the sharp edges of the window.
- The spectrum falls off relatively slowly as the frequency f_k moves away from f_0 . This happens because of the sharp edges on the rectangular window function. The slow fall-off is referred to as *spectral leakage*. It is a significant problem if you're analyzing spectra that contain both very strong and very weak frequency components, since the weak frequency components can be hidden by the spectral leakage of the strong frequency components.

To mitigate these issues, other window functions are often used. One of the most common is the *Hanning window*

$$W(n) = \sin^2\left(\frac{\pi n}{N}\right), \quad 0 < n < N-1 \quad (17)$$

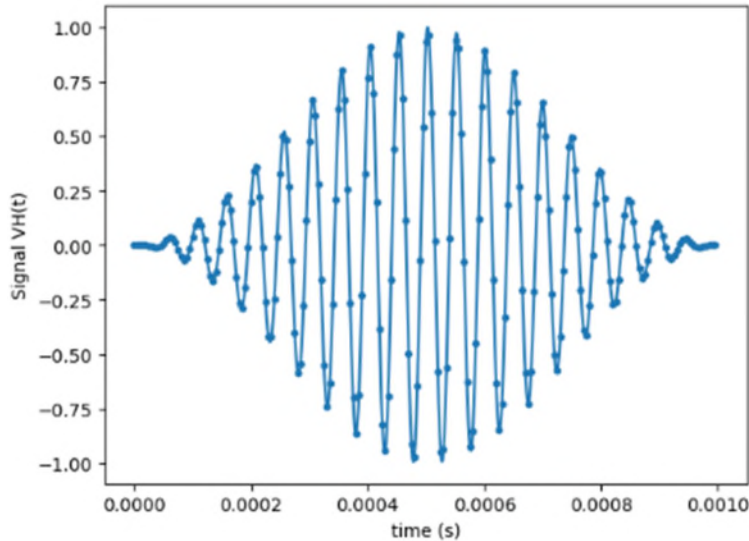


Figure 14. Input signal example of Section II.C with the Hanning window applied.

Figure 14 shows what our input signal example looks like with the Hanning window applied. Figure 15 shows a comparison of the power spectrum for the rectangular window to the power spectrum with the Hanning window.

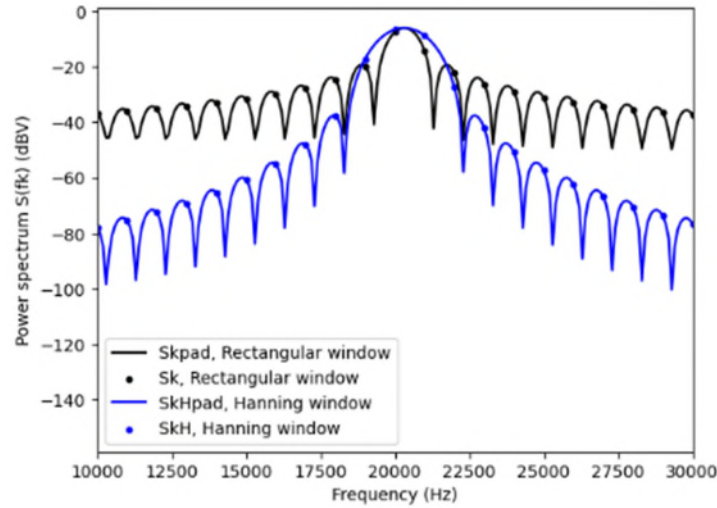


Figure 15. Comparison of the power spectrum for a rectangular window with the power spectrum for a Hanning window for the signal example of section II.C.

The Hanning window has one disadvantage relative to the rectangular window, which is that the central part of the peak is a bit wider. However it has the following advantages over the rectangular window:

- Since the central part of the peak is wider, the value of the highest point in the power spectrum will generally be close to the true peak value of the power.
- Since there are no sharp edges in the window, the exact shape of the power spectrum will be much less sensitive to exactly how the signal is cut-off by the window.
- The power spectrum falls much more rapidly as the frequency f_k moves away from the peak. This decreases problems in distinguishing frequency components of signals caused by spectral leakage.

The Hanning window provides a pretty good compromise between minimizing spectral leakage and avoiding too much broadening of the central peak, and is often used in practice.

I. Example Oscilloscope FFT display

To see how all of this works in practice, let's look at the example of an oscilloscope FFT display shown in Figure 22. This is a display from a Tektronix TBS 2000B series oscilloscope.

Here are some of the main points to notice about this display:

- The oscilloscope displays the time-domain signal $V(t)$ in the upper part of the display, and the corresponding power spectrum $S(f_k)$ in the lower part. Since the signal is approximately a square wave, the power spectrum shows a peak at the frequency f_0 of the square wave and at a series of odd harmonics of f_0 .

- The spectrum is shown *single-sided*.
- The spectrum is shown on a dBV scale at 20 dB per division. (According to the oscilloscope manual, dB on the display stands for “dBV-rms.”)
- $N = 20,000$ and the sampling rate is $f_s = 500,000$ S/s. From this we can determine that

$$\Delta t = \frac{1}{f_s} = 2 \text{ ns}, \text{ and that the spacing between adjacent frequency points is}$$

$$\Delta f = \frac{1}{N\Delta t} = 25 \text{ kHz. The full span of the frequency span is}$$

$$25 \text{ MHz/div} \times 10 \text{ div} = 250 \text{ MHz} = f_s / 2, \text{ as expected.}$$

- A Hanning window has been applied to the data before the FFT is taken. The peaks appear very sharp because the data contains so many points. Because of this, the frequency resolution of the spectrum $\sim$ few times Δf is very high. We would need to magnify the horizontal scale to see the structure of an individual peak.



Figure 16. Example of an oscilloscope FFT display. Figure is from the [TBS2000B series data sheet](#)